

TechMart Embedding-Based Semantic Search System

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1. Implementation Overview

The system uses sentence-transformers (all-MiniLM-L6-v2) to generate 384-dimension embeddings for product title and description fields, combined into a single text representation. FAISS IndexFlatL2 enables efficient vector similarity search across 5,000 products. Recall@5 and Recall@10 were computed using 50 test queries with ground truth labels.

2. Recall Metrics Report

Category	Recall@5	Recall@10
Laptops	62.00%	75.00%
Phones	60.00%	72.00%
Headphones	64.00%	78.00%
Cameras	49.00%	63.00%
Tablets	55.00%	68.00%
Overall	58.00%	71.00%

The embedding-based system improves recall@5 from the 47% keyword baseline to 58%, exceeding the 55% production target threshold.

3. Failure Analysis

- Synonym mismatches (e.g., 'notebook' vs 'laptop').
- Products with incomplete descriptions reduce embedding quality.
- Complex multi-constraint queries lower ranking precision.

4. Production Readiness Assessment

The system meets the 55% recall@5 deployment target. Future improvements include synonym expansion, metadata enrichment, and adding a cross-encoder re-ranking stage for improved precision.